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Inventor(s)	Hu; Lijiao et al.

Detection chip, method for using detection chip, and detection device

Abstract

A detection chip, a method for using a dejection chip, and a detection device are provided. The detection chip includes a chip substrate and a first sealing film that are stacked. The chip substrate includes a first surface, and the first sealing film covers the first surface of the chip substrate. The chip substrate further includes a fluid channel on the first surface, and the fluid channel includes a plurality of membrane valve portions. The membrane valve portions are configured to allow a portion of fee first sealing film covering the membrane valve portions to approach and separate, so as to close and open the fluid channel correspondingly.

Inventors:	Hu; Lijiao (Beijing, CN), Cui; Haochen (Beijing, CN), Yuan; Chungeng (Beijing, CN), Hu; Tao (Beijing, CN), Li; Jing (Beijing, CN), Gan; Weiqiong (Beijing, CN), Shen; Xiaohe (Beijing, CN)
Applicant:	BEIJING BOE HEALTH TECHNOLOGY CO., LTD. (Beijing, CN); BOE TECHNOLOGY GROUP CO., LTD. (Beijing, CN)
Family ID:	1000008778932
Assignee:	BEIJING BOE HEALTH TECHNOLOGY CO., LTD. (Beijing, CN); BOE TECHNOLOGY GROUP CO., LTD. (Beijing, CN)
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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0266582	12/2004	Modlin	436/164	G01N 21/77
2010/0291588	12/2009	McDevitt	435/287.1	G01N 33/491
2011/0162785	12/2010	Zhou et al.	N/A	N/A
2012/0178091	12/2011	Glezer	435/6.12	B01L 3/5027
2015/0093838	12/2014	Landers et al.	N/A	N/A
2016/0186240	12/2015	Andreyev	435/287.2	B01L 3/5029
2018/0093269	12/2017	Rachamim	N/A	B01L 3/502738
2022/0032291	12/2021	Wu	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105349401	12/2015	CN	N/A
205127987	12/2015	CN	N/A
101750488	12/2015	CN	N/A
101750489	12/2015	CN	N/A
108816300	12/2017	CN	N/A

Primary Examiner: Wecker; Jennifer*Assistant Examiner:* Washington; Britney N.

Background/Summary

(1) This application is a U.S. National Phase Entry of International Application No. PCT/CN2021/074635 filed on Feb. 1, 2021, designating the United States of America and claiming priority to Chinese Patent Application No. 202010104016.2, filed on Feb. 20, 2020. The present application claims priority to and the benefit of the above-identified applications and the above-identified applications are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(2) Embodiments of the present disclosure relate to a detection chip, a method for using a detection chip, and a detection device.

BACKGROUND

(3) Microfluidic chip technology integrates basic operation units such as sample preparation, reaction, separation and detection involved in the fields of biology, chemistry, medicine, or the like, into a chip with a micrometer-scale micro channel, so as to automatically complete the whole process of reaction and analysis. The chip used in this process is referred to as a microfluidic chip, and may also be referred to as lab-on-a-chip. Microfluidic chip technology has the advantages of small sample consumption, fast analysis speed, easy to be made into portable instruments, suitable for instant and on-site analysis, or the like, and has been widely used in various fields such as biology, chemistry, medicine, and the like.

SUMMARY

(4) At least one embodiment of the present disclosure provides a detection chip, which comprises a chip substrate and a first sealing film that are stacked. The chip substrate comprises a first surface, and the first sealing film covers the first surface of the chip substrate; the chip substrate further comprises a fluid channel on the first surface, and the fluid channel comprises a plurality of membrane valve portions; and the membrane valve portions are configured to allow a portion of the first sealing film covering the membrane valve portions to approach and separate, so as to close and open the fluid channel correspondingly.

(5) For example, the detection chip provided by an embodiment of the present disclosure further comprises a membrane valve sealing plate. The membrane valve sealing plate is on a side of the first sealing film away from the chip substrate and comprises a plurality of protruding structures, the plurality of protruding structures and the plurality of membrane valve portions are in one-to-one correspondence, and in the case where the plurality of protruding structures are in contact with the plurality of membrane valve portions, respectively, the fluid channel is closed.

(6) For example, in the detection chip provided by an embodiment of the present disclosure, the first sealing film is an elastic film.

(7) For example, in the detection chip provided by an embodiment of the present disclosure, the chip substrate further comprises at least one liquid cell, the at least one liquid cell is communicated with the fluid channel, and at least one of the plurality of membrane valve portions is configured to close and open a portion of the fluid channel communicated with the at least one liquid cell.

(8) For example, in the detection chip provided by an embodiment of the present disclosure, the fluid channel further comprises an extraction region and a plurality of first branches, and the at least one liquid cell comprises a plurality of liquid cells; the plurality of first branches are communicated with the plurality of liquid cells in one-to-one correspondence, and each of the plurality of first branches is communicated with the extraction region; and the plurality of membrane valve portions comprise a plurality of first membrane valve portions respectively on the plurality of first branches, so as to control the plurality of first branches to be opened or closed.

(9) For example, in the detection chip provided by an embodiment of the present disclosure, the fluid channel further comprises a reaction region and a plurality of second branches; the reaction region is communicated with the extraction region and at least one of the plurality of liquid cells through the plurality of second branches, respectively; and the plurality of membrane valve portions further comprise a plurality of second membrane valve portions respectively on the plurality of second branches, so as to control the plurality of second branches to be opened or closed.

(10) For example, in the detection chip provided by an embodiment of the present disclosure, the reaction region comprises a porous structure, the porous structure comprises a plurality of liquid containing holes, and the plurality of liquid containing holes are configured to contain identical or different amplification primers.

(11) For example, in the detection chip provided by an embodiment of the present disclosure, the plurality of liquid cells comprise a first liquid cell, a second liquid cell, a third liquid cell, a fourth liquid cell, and a fifth liquid cell, the first liquid cell is configured to contain a lysis solution, the second liquid cell is configured to contain a first rinsing solution, the third liquid cell is configured to contain a second rinsing solution, the fourth liquid cell is configured to contain an eluent, and the fifth liquid cell is configured to accommodate a waste liquid generated in the extraction region during a reaction process.

(12) For example, the detection chip provided by an embodiment of the present disclosure further comprises a second sealing film. The chip substrate comprises a second surface opposite to the first surface, and the second sealing film covers the second surface of the chip substrate.

(13) For example, in the detection chip provided by an embodiment of the present disclosure, the second sealing film is a composite film comprising metal foil and a polymer material that are stacked.

(14) For example, the detection chip provided by an embodiment of the present disclosure further comprises an adhesive layer. The adhesive layer is between the chip substrate and the first sealing film, and is configured to allow the chip substrate to be adhered with the first sealing film, and the adhesive layer exposes the fluid channel of the chip substrate.

(15) At least one embodiment of the present disclosure further provides a detection device, which is adapted to operate the detection chip provided by any one of the embodiments of the present disclosure. The detection device comprises a membrane valve control unit, the membrane valve control unit is configured to mount the detection chip and comprises at least one protruding portion, and the at least one protruding portion is movable to control the portion of the first sealing film covering the membrane valve portions whether to approach the membrane valve portions or whether to separate from the membrane valve portions in the case where the detection chip is mounted on the membrane valve control unit, so as to close and open the fluid channel correspondingly.

(16) For example, the detection device provided by an embodiment of the present disclosure further comprises a film driving unit. In the case where the fluid channel comprises an extraction region, the film driving unit is configured to, in the case where the detection chip is mounted on the membrane valve control unit, apply pressure to a portion of the first sealing film covering the extraction region, so as to allow the portion of the first sealing film covering the extraction region to deform.

(17) At least one embodiment of the present disclosure further provides a method for using the detection chip provided by any one of the embodiments of the present disclosure. The method comprises: controlling the plurality of membrane valve portions to allow the portion of the first sealing film covering the plurality of membrane valve portions to be separated from the plurality of membrane valve portions, respectively, so as to open the fluid channel correspondingly.

(18) For example, in the method provided by an embodiment of the present disclosure, in the case where the plurality of membrane valve portions comprise a plurality of first membrane valve

portions and a plurality of second membrane valve portions, the chip substrate comprises a plurality of liquid cells, and the fluid channel comprises an extraction region and a reaction region, controlling the plurality of membrane valve portions to allow the portion of the first sealing film covering the plurality of membrane valve portions to be separated from the plurality of membrane valve portions, respectively, so as to open the fluid channel correspondingly, comprises: controlling the plurality of first membrane valve portions to allow the plurality of liquid cells to be communicated with the extraction region, so as to allow liquid in the plurality of liquid cells to enter the extraction region; and controlling the plurality of second membrane valve portions to allow the reaction region to be communicated with the extraction region, so as to allow liquid in the extraction region to enter the reaction region.

(19) For example, the method provided by an embodiment of the present disclosure further comprises: controlling flow of liquid in the detection chip by applying pressure to a portion of the first sealing film covering the extraction region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to clearly illustrate the technical solutions of the embodiments of the present disclosure, the drawings of the embodiments will be briefly described in the following. It is obvious that the described drawings are only related to some embodiments of the present disclosure and thus are not limitative to the present disclosure.

(2) FIG. 1 is a perspective exploded view of a three-dimensional structure of a detection chip provided by at least one embodiment of the present disclosure;

(3) FIG. 2 is a non-perspective exploded view of the three-dimensional structure of the detection chip illustrated in FIG. 1;

(4) FIG. 3 is a perspective view of the three-dimensional structure of the detection chip illustrated in FIG. 1;

(5) FIG. 4 is a side perspective view of the detection chip illustrated in FIG. 1;

(6) FIG. 5 is a top perspective view of the detection chip illustrated in FIG. 1;

(7) FIG. 6 is a partial enlarged perspective view of a reaction region of a detection chip provided by at least one embodiment of the present disclosure;

(8) FIG. 7A is a schematic block diagram of a detection device provided by at least one embodiment of the present disclosure;

(9) FIG. 7B is a schematic structural diagram of a detection device provided by at least one embodiment of the present disclosure;

(10) FIG. 8 is a schematic block diagram of another detection device provided by at least one embodiment of the present disclosure;

(11) FIG. 9 is a schematic flowchart of a method for using a detection chip provided by at least one embodiment of the present disclosure;

(12) FIG. 10 is a schematic flowchart of step S10 of the method for using the detection chip illustrated in FIG. 9; and

(13) FIG. 11 is a schematic flowchart of another method for using the detection chip provided by at least one embodiment of the present disclosure.

DETAILED DESCRIPTION

(14) In order to make objects, technical details and advantages of the embodiments of the present disclosure apparent, the technical solutions of the embodiments will be described in a clearly and fully understandable way in connection with the drawings related to the embodiments of the present disclosure. Apparently, the described embodiments are just a part but not all of the embodiments of the present disclosure. Based on the described embodiments herein, those skilled

in the art can obtain other embodiment(s), without any inventive work, which should be within the scope of the present disclosure.

(15) Unless otherwise defined, all the technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which the present disclosure belongs. The terms “first,” “second,” etc., which are used in the description and the claims of the present application for disclosure, are not intended to indicate any sequence, amount or importance, but distinguish various components. Also, the terms such as “a,” “an,” etc., are not intended to limit the amount, but indicate the existence of at least one. The terms “comprise,” “comprising,” “include,” “including,” etc., are intended to specify that the elements or the objects stated before these terms encompass the elements or the objects and equivalents thereof listed after these terms, but do not preclude the other elements or objects. The phrases “connect,” “connected,” “coupled,” etc., are not intended to define a physical connection or mechanical connection, but may include an electrical connection, directly or indirectly. “On,” “under,” “right,” “left” and the like are only used to indicate relative position relationship, and when the position of the object which is described is changed, the relative position relationship may be changed accordingly.

(16) In the design process of microfluidic chips, it is usually desirable to integrate as many functions of analysis and detection on the chip as possible to reduce the dependence of the chip on external operations, thereby achieving automation and integration. Microfluidic chips are mostly disposable products, which can omit the complicated liquid path system for cleaning, waste liquid treatment, or the like, and can further avoid pollution caused by the liquid path system. In order to achieve integration, the reagent storage component may be provided in the microfluidic chip to store various reagents required for analysis and detection. For the general microfluidic chip with the reagent storage function, the structure of the chip is relatively complicated, or the preparation process is relatively complicated, which causes the high cost of the microfluidic chip as a consumable, and the delivery of the reagent cannot be accurately and precisely controlled. Meanwhile, the process of the microfluidic chip that can realize multiple detection is more complicated, and the cost is relatively high.

(17) At least one embodiment of the present disclosure provides a detection chip, a method for using a detection chip, and a detection device. The detection chip has a simple structure and can quantitatively deliver the reagent. Further, at least one example thereof can further realize multiple detection. Furthermore, at least one example thereof can further facilitate improving the heat conduction efficiency and the stability and accuracy of optical detection, and can effectively prevent the reagent leakage during transportation.

(18) Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the drawings. It should be noted that the same reference numerals in different drawings are used to refer to the same described elements.

(19) At least one embodiment of the present disclosure provides a detection chip, and the detection chip includes a chip substrate and a first sealing film which are stacked. The chip substrate includes a first surface, and the first sealing film covers the first surface of the chip substrate. The chip substrate further includes a fluid channel on the first surface, and the fluid channel includes a plurality of membrane valve portions. The membrane valve portions are configured to allow a portion, covering the membrane valve portions, of the first sealing film to approach and separate, so as to close and open the fluid channel correspondingly, thereby allowing quantitative delivery of the reagent.

(20) FIG. 1 is a perspective exploded view of a three-dimensional structure of a detection chip provided by at least one embodiment of the present disclosure, FIG. 2 is a non-perspective exploded view of the three-dimensional structure of the detection chip illustrated in FIG. 1, FIG. 3 is a perspective view of the three-dimensional structure of the detection chip illustrated in FIG. 1, FIG. 4 is a side perspective view of the detection chip illustrated in FIG. 1, and FIG. 5 is a top perspective view of the detection chip illustrated in FIG. 1.

(21) Hereinafter, the detection chip provided by some embodiments of the present disclosure is described with reference to FIG. 1 to FIG. 5.

(22) As illustrated in FIG. 1 to FIG. 3, the detection chip **100** includes a chip substrate **10** and a first sealing film **20** which are stacked.

(23) The chip substrate **10** includes a first surface **11** and a fluid channel **12** located on the first surface **11**. For example, the first surface **11** is the lower surface of the chip substrate **10** illustrated in FIG. 1 to FIG. 3, and the fluid channel **12** is located on the lower surface of the chip substrate **10**. For example, the material of the chip substrate **10** is polypropylene (PP), and the chip substrate **10** is processed by an injection molding process. By designing a corresponding injection mould, the fluid channel **12** can be formed on the first surface **11** of the chip substrate **10** in a recessed manner. Certainly, the embodiments of the present disclosure are not limited to this, and any applicable process, such as laser engraving, photolithography, or the like, may also be used to fabricate the fluid channel **12**. It should be noted that in the embodiments of the present disclosure, the material and processing method of the chip substrate **10** are not limited, which can be determined according to actual requirements.

(24) For example, the first sealing film **20** covers the first surface **11** of the chip substrate **10**. Because the fluid channel **12** is provided on the first surface **11** of the chip substrate **10** in the form of recess, a space for liquid (for example, various reagents required for analysis and detection) flowing can be formed between the first sealing film **20** and the fluid channel **12**. For example, a space for reagent reaction can also be formed. For example, the first sealing film **20** is an elastic film, such as an elastic transparent film. For example, the material of the first sealing film **20** is polyethylene terephthalate (PET) to have better elasticity and strength, so that the first sealing film **20** can be restored to the original state after elastic deformation. Certainly, the embodiments of the present disclosure are not limited to this, and the first sealing film **20** may also be made of other applicable materials, such as a polymer composite material of polystyrene (PS) and PET, so as to have better elasticity and strength.

(25) As illustrated in FIG. 5, the fluid channel **12** includes a plurality of membrane valve portions **13**, an extraction region **121**, a plurality of first branches **122**, a reaction region **123**, and a plurality of second branches **124**.

(26) The membrane valve portion **13** is configured to allow the portion of the first sealing film **20** covering the membrane valve portion **13** to approach and separate, so that the fluid channel **12** can be closed and opened correspondingly. For example, under the action (for example, pressing) of a separately provided component, the portion of the first sealing film **20** covering the membrane valve portion **13** is squeezed and deformed, such as elastically deformed, so as to be close to the chip substrate **10** (for example, to be completely attached with the chip substrate **10**), so that the space between the first sealing film **20** and the fluid channel **12** is reduced or even cut off at the membrane valve portion **13**, and the liquid cannot pass through the membrane valve portion **13**, so as to close the fluid channel **12** accordingly. For example, under the action (for example, loosening) of a separately provided component, the portion of the first sealing film **20** covering the membrane valve portion **13** and being contacted with the chip substrate **10** is recovered, thereby being separated from the chip substrate **10**, so that the space between the first sealing film **20** and the fluid channel **12** is restored at the membrane valve portion **13**, and the liquid can pass through the membrane valve portion **13**, so as to correspondingly open the fluid channel **12**.

(27) For example, the membrane valve portion **13** is a circular recess as illustrated in FIG. 5. Accordingly, the separately provided component for controlling the membrane valve portion **13** is a cylindrical protrusion, so that the membrane valve portion **13** can be squeezed. Certainly, the embodiments of the present disclosure are not limited to this, the membrane valve portion **13** may also be in any other applicable shapes, such as a rectangle, a hexagon, an ellipse, etc., and correspondingly, the separately provided component for controlling the membrane valve portion **13** may be a columnar protrusion with a cross-sectional shape of, e.g., a rectangle, a hexagon, an

ellipse, etc., so that the membrane valve portion **13** can be squeezed.

(28) For example, the chip substrate **10** further includes at least one liquid cell **14**, and the at least one liquid cell **14** is communicated with the fluid channel **12**. For example, in some examples, as illustrated in FIG. **1** and FIG. **5**, the at least one liquid cell **14** includes a plurality of liquid cells, for example, five liquid cells, namely, a first liquid cell **141**, a second liquid cell **142**, a third liquid cell **143**, a fourth liquid cell **144**, and a fifth liquid cell **145**. The first liquid cell **141** is configured to contain a lysis solution, the second liquid cell **142** is configured to contain a first rinsing liquid, the third liquid cell **143** is configured to contain a second rinsing liquid, the fourth liquid cell **144** is configured to contain an eluent, and the fifth liquid cell **145** is configured to accommodate a waste liquid generated in the extraction region **121** during the reaction process.

(29) For example, at least one of the plurality of membrane valve portions **13** is configured to close and open the portion of the fluid channel **12** communicated with at least one liquid cell **14**. For example, in some examples, as illustrated in FIG. **1** and FIG. **5**, the plurality of first branches **122** are communicated with the plurality of liquid cells **14** in one-to-one correspondence, and each of the plurality of first branches **122** is communicated with the extraction region **121**. The plurality of membrane valve portions **13** include a plurality of first membrane valve portions **131** to **135** respectively located on the plurality of first branches **122**, so as to control the plurality of first branches **122** to be opened or closed. The liquid stored in the plurality of liquid cells **14** can enter the extraction region **121** along the first branch **122**, so as to facilitate operations such as extraction, rinsing, elution, or the like in the extraction region **121**.

(30) For example, the extraction region **121** includes a plurality of magnetic beads **001**, and the plurality of magnetic beads **001** are movably distributed in the extraction region **121**. For example, the surface of the magnetic bead **001** is processed with modified treatment. In the case where the detection chip **100** is used for detection, for example, in the case where the detection chip **100** is used to detect specific nucleic acid fragments, the magnetic bead **001** can allow molecular structures such as the nucleic acid fragments to be bonded to the magnetic bead **001** during the extraction process when detection is performed, so as to achieve the function of extraction. For example, the molecular structures such as the aforementioned nucleic acid fragments are obtained after the sample to be detected is split. The description of modifying the surface of the magnetic bead **001** may refer to the conventional design, which will not be described in detail here.

(31) For example, the reaction region **123** is communicated with the extraction region **121** and at least one of the plurality of liquid cells **14** (for example, communicated with the fifth liquid cell **145**) through the plurality of second branches **124**, respectively. The plurality of membrane valve portions **13** further include a plurality of second membrane valve portions **136** to **137** respectively located on the plurality of second branches **124**, so as to control the plurality of second branches **124** to be open or closed. For example, the reaction region **123** can contain the reaction liquid after operations such as extraction, rinsing, elution, etc., and allow the reaction liquid to perform the amplification reaction and to be subjected with subsequent optical detection in the reaction region **123**. For example, in the case where the reaction region **123** and the extraction region **121** are communicated with each other, the reaction region **123** is communicated with the fifth liquid cell **145**, so that the fifth liquid cell **145** can function as a vent hole, thereby facilitating allowing the reaction liquid to enter the reaction region **123** from the extraction region **121**. For example, when the reaction liquid enters the reaction region **123**, the pressure in the reaction region **123** increases, and the excess air in the reaction region **123** can be discharged to the fifth liquid cell **145** through the second branch **124**, so as to balance the air pressure and facilitate allowing the reaction liquid to enter the reaction region **123**.

(32) Thus, by providing the first branches **122** and the second branches **124**, and correspondingly providing the first membrane valve portions **131** to **135** and the second membrane valve portions **136** to **137**, each liquid cell **14** can be individually controlled whether to be communicated with the extraction region **121**, and the reaction region **123** can be controlled whether to be communicated

with the extraction region **121**, so that the detection chip **100** can be operated to realize the function of the detection chip **100**.

(33) In the embodiments of the present disclosure, the membrane valve portion **13** can control whether the liquid in the fluid channel **12** passes or not, and can be used as a sealed valve of the liquid cell **14** to control when the liquid cell **14** is opened to release the reagent therein. Because the amount of the reagent passed is basically fixed during the membrane valve portion **13** being opened once, the membrane valve portion **13** can also quantitatively deliver the reagent to realize microliter-level liquid transmission.

(34) It should be noted that although the extraction region **121** and the reaction region **123** illustrated in FIG. 5 are circular depressions, this does not constitute a limitation to the embodiments of the present disclosure. The extraction region **121** and the reaction region **123** may also be depressions in any other applicable shape, such as a rectangle, a hexagon, an ellipse, etc., as long as a space for accommodating the liquid can be formed, and the embodiments of the present disclosure are not limited in this aspect.

(35) The respective sizes of the membrane valve portion **13**, the extraction region **121**, the first branch **122**, the reaction region **123**, and the second branch **124** are not limited, which may be determined according to actual needs, as long as the membrane valve portion **13** can be ensured to control the first branch **122** and the second branch **124** to be opened and closed.

(36) It should be noted that, in the embodiments of the present disclosure, the first sealing film **20** is, for example, an elastic transparent plastic thin film (such as a PET film), and the first sealing film **20** has certain elasticity and strength, and is pushed up and down and pulled up and down after applying positive and negative pressure (for example, positive and negative air pressure) to the portion of the first sealing film **20** covering the extraction region **121**. Therefore, in the case where the fluid channel **12** is not closed, the liquid can be pumped quantitatively, thereby controlling the flow of the liquid between each liquid cell **14**, the extraction region **121**, and the reaction region **123**. Because the first sealing film **20** is thin and can achieve rapid heat conduction, heat can be transferred quickly when the reaction liquid in the reaction region **123** is heated, which can facilitate improving the heat conduction efficiency and speeding up the amplification reaction. The first sealing film **20** is a transparent thin film, so that when performing optical detection on the liquid in the reaction region **123** that completes the amplification reaction, the light transmittance is higher, which facilitates improving the stability and accuracy of the optical detection.

(37) For example, as illustrated in FIG. 1 to FIG. 4, in at least one embodiment of the present disclosure, the detection chip **100** may further include a membrane valve sealing plate **30**. The membrane valve sealing plate **30** is provided on a side of the first sealing film **20** away from the chip substrate **10**, and for example, the membrane valve sealing plate **30** is provided adjacent to the first sealing film **20**. The membrane valve sealing plate **30** includes a plurality of protruding structures **31**. The plurality of protruding structures **31** and the plurality of membrane valve portions **13** are in one-to-one correspondence, and in the case where the plurality of protruding structures **31** are respectively in contact with the plurality of membrane valve portions **13**, the fluid channel **12** is closed.

(38) For example, the plurality of protruding structures **31** include seven protruding structures **311** to **317**, and correspondingly, the plurality of membrane valve portions **13** also include seven membrane valve portions **131** to **137**. The seven protruding structures **311** to **317** are correspondingly distributed according to the distribution positions of the seven membrane valve portions **131** to **137**, so that each protruding structure **31** can be correspondingly inserted into each membrane valve portion **13** at the same time. Thus, in the case where each protruding structure **31** and each membrane valve portion **13** are correspondingly in contact with each other (as illustrated in FIG. 4), the portion of the first sealing film **20** covering the membrane valve portion **13** is squeezed and deformed to be completely attached to the chip substrate **10**, thereby allowing the fluid channel **12** to be closed.

(39) For example, because the membrane valve portion **13** is a circular depression as illustrated in FIG. **5**, the protruding structure **31** is a cylindrical protrusion. When the shape of the membrane valve portion **13** changes, in order to achieve a better matching effect, the shape of the protruding structure **31** needs to be changed accordingly.

(40) For example, the membrane valve sealing plate **30** can be fixed on the chip substrate **10** by screw connection (for example, a screw **32**), clamping, and other fixing methods, and the first sealing film **20** is located between the membrane valve sealing plate **30** and the chip substrate **10**. For example, the fixing method is a detachable fixing method. For example, during transportation or before the detection chip **100** is used, the membrane valve sealing plate **30** is fixed on the chip substrate **10**, so that the fluid channel **12** can be closed, so as to enable the liquid in each liquid cell **14** not to leak or mix. When the detection chip **100** is used, the membrane valve sealing plate **30** is separated from the chip substrate **10**, and a separately provided device (such as a detection device including a plurality of cylindrical protrusions which can be independently controlled) is used to control each membrane valve portion **13**, thereby achieving the function of the detection chip **100**.

(41) For example, the material of the membrane valve sealing plate **30** may be acrylonitrile butadiene styrene (ABS) plastic, or other applicable materials, which are not limited in the embodiments of the present disclosure.

(42) For example, as illustrated in FIG. **1** to FIG. **4**, in at least one embodiment of the present disclosure, the detection chip **100** may further include a second sealing film **40**. The chip substrate **10** includes a second surface **15** opposite to the first surface **11**, and the second sealing film **40** covers the second surface **15** of the chip substrate **10**. For example, the second surface **15** is the upper surface of the chip substrate **10** illustrated in FIG. **1** to FIG. **4**, and the second sealing film **40** covers the upper surface of the chip substrate **10**. The second sealing film **40** can be provided to cooperate with the membrane valve portion **13** to seal the liquid in the liquid cell **14** within the liquid cell **14**, so as to prevent the liquid in the liquid cell **14** from leaking during transportation.

(43) For example, the second sealing film **40** is a composite film including metal foil and a polymer material which are stacked, such as a composite film of aluminum foil and a polymer material, so as to facilitate the bonding with the chip substrate **10** by heat pressing, and further facilitate being punctured when the sample needs to be added. For example, as illustrated in FIG. **4**, when the detection chip **100** is used, a separately provided piercing mechanism **401** (for example, any applicable hard object) may be used to pierce the second sealing film **40**, so as to add the sample to the first liquid cell **141** through the puncture on the second sealing film **40**. For example, in some other examples, the detection chip **100** may also include the piercing mechanism **401** and further include a piercing mechanism fixing plate **402**. The piercing mechanism fixing plate **402** has openings **403** corresponding to the piercing mechanism **401**, the piercing mechanism **401** is disposed in the opening **403**, and the piercing mechanism **401** can move along the axial direction of the opening **403** in the opening **403**.

(44) For example, as illustrated in FIG. **1** to FIG. **4**, in at least one embodiment of the present disclosure, the detection chip **100** may further include an adhesive layer **50**. The adhesive layer **50** is provided between the chip substrate **10** and the first sealing film **20**, and is configured to adhere the chip substrate **10** with the first sealing film **20**. For example, the adhesive layer **50** may include an adhesive material such as an acrylic adhesive, or the like, and for example, may be implemented as a double-sided tape. For example, the chip substrate **10**, the adhesive layer **50**, and the first sealing film **20** have substantially the same contours, so that the adhesive layer **50** can achieve a stronger bond between the chip substrate **10** and the first sealing film **20**.

(45) For example, the adhesive layer **50** exposes the fluid channel **12** of the chip substrate **10**, that is, the adhesive layer **50** includes a hollow region **51**, and the shape of the hollow region **51** is the same as or substantially the same as the orthographic projection of the fluid channel **12** on the adhesive layer **50**, so as to facilitate forming a space for liquid flow and reagent reaction between the first sealing film **20** and the fluid channel **12**.

(46) For example, in some other examples, in the case where the first sealing film **20** is bonded to the chip substrate **10** by ultrasonic welding, photosensitive adhesive bonding, chemical solvent bonding, laser welding, or the like, the adhesive layer **50** may be omitted.

(47) FIG. **6** is a partial enlarged perspective view of a reaction region of a detection chip provided by at least one embodiment of the present disclosure. For example, as illustrated in FIG. **6**, the reaction region **123** includes a porous structure **125**, the porous structure **125** includes a plurality of liquid containing holes **002**, and the plurality of liquid containing holes **002** are configured to contain the same or different amplification primers. For example, the amplification primer is a lyophilized reagent, and the reaction liquid entering the reaction region **123** can re-thaw the lyophilized reagent, and the desired reaction (such as an amplification reaction) occurs, so that the optical detection can be performed after the reaction is completed. In the case where the liquid containing holes **002** store different amplification primers, the reaction liquid entering respective liquid containing holes **002** may undergo different amplification reactions (that is, the amplified objects are different), so that multiple objects can be detected (such as different types of viruses) to achieve multiple detection. Because the amplification primer is the lyophilized reagent, the amplification primer stored in different liquid containing hole **002** may not be mixed with each other during transportation, and may not be moved out of the liquid containing hole **002**. It should be noted that, in the embodiments of the present disclosure, the cross-sectional shape, number, and distribution method of the liquid containing holes **002** are not limited, which can be determined according to actual requirements.

(48) The working principle of the detection chip **100** is exemplified below.

(49) In the production process, the lysis solution is pre-embedded in the first liquid cell **141**, the first rinsing liquid is pre-embedded in the second liquid cell **142**, the second rinsing liquid is pre-embedded in the third liquid cell **143**, the eluent is pre-embedded in the fourth liquid cell **144**, the fifth liquid cell **145** is vacant, and the amplification primer is pre-embedded in the liquid containing holes **002** of the reaction region **123**. The membrane valve sealing plate **30** is mounted on the chip substrate **10**, so that the plurality of protruding structures **31** are in contact with the plurality of membrane valve portions **13**, respectively, thereby allowing the fluid channel **12** to be closed and sealing the liquid of each liquid cell **14** within the corresponding liquid cell **14**. For example, taking it as an example that the human papillomavirus is the sample to be detected, the components of the lysis solution are guanidine hydrochloride, 3-(N-morpholine) propanesulfonic acid (MOPS), and a mixture of polyoxyethylene sorbitan monolaurate and polyoxyethylene dihydrosorbitol monolaurate (Tween), the first rinsing liquid consists of guanidine hydrochloride, MOPS, and isopropanol, the second rinsing liquid consists of guanidine hydrochloride, MOPS, and ethanol, and the eluent consists of trihydroxymethyl aminomethane (Tris) and ethylene diamine tetraacetic acid (EDTA).

(50) During use, the membrane valve sealing plate **30** is separated from the chip substrate **10**, and the detection chip **100** is mounted on a separately provided detection device. For example, the detection device includes a plurality of protruding portions, the plurality of protruding portions and the plurality of membrane valve portions **13** are in one-to-one correspondence, and each membrane valve portion **13** can be individually controlled.

(51) First, the portion of the second sealing film **40** covering the first liquid cell **141** is pierced, and the sample to be detected is added to the first liquid cell **141**. For example, any suitable hard object can be used to pierce the second sealing film **40**. The sample to be detected is, for example, blood, body fluid, etc., which is not limited in the embodiments of the present disclosure. The sample to be detected is lysed (for example, the lysis temperature range may be determined according to actual requirements) under the action of the lysis solution in the first liquid cell **141**, thereby being lysed to obtain nucleic acid fragments. The protruding portion in the detection device is controlled to open the first membrane valve portion **133**, and the detection device is used to apply low-frequency positive and negative air pressure (or only apply negative air pressure or positive air

pressure depending on the actual situation) to the portion of the first sealing film **20** covering the extraction region **121**, so as to drive the liquid in the first liquid cell **141** to flow into the extraction region **121**. Then, the first membrane valve portion **133** is closed. The high-frequency positive and negative air pressure is applied to the portion of the first sealing film **20** covering the extraction region **121**, so that the portion of the first sealing film **20** covering the extraction region **121** vibrates repeatedly, so that the liquid in the extraction region **121** vibrates, which facilitates the bonding of the magnetic bead **001** pre-embedded in the extraction region **121** with the nucleic acid fragments in the liquid, thereby achieving the extraction of the nucleic acid fragments.

(52) Then, the protruding portion in the detection device is controlled to open the first membrane valve portion **135**, and the above-mentioned air pressure applying method is used to apply the air pressure to the portion of the first sealing film **20** covering the extraction region **121**, so that the first rinsing liquid pre-embedded in the second liquid cell **142** is injected into the extraction region **121**. Next, the first membrane valve portion **135** is closed, and high-frequency positive and negative air pressure is applied to the portion of the first sealing film **20** covering the extraction region **121**, so that the portion of the first sealing film **20** covering the extraction region **121** vibrates repeatedly, thereby allowing the liquid in the extraction region **121** to vibrate, so as to wash away the protein. Then, the first membrane valve portion **134** is opened, and the magnet in the detection device is used to attract the magnetic beads **001** in the extraction region **121** (for example, the magnet is moved up to be close to the portion of the first sealing film **20** that covers the extraction region **121**). Air pressure is applied to the first sealing film **20** to drive the liquid in the extraction region **121** to flow into the fifth liquid cell **145**. At this time, because the magnetic beads **001** are fixed in the extraction region **121** under the attractive force of the magnet, the nucleic acid fragments adsorbed on the magnetic beads **001** may not enter the fifth liquid cell **145** along with the liquid. For example, the fifth liquid cell **145** serves as a waste liquid container for containing the waste liquid generated in the extraction region **121**. Then, the first membrane valve portion **134** is closed and the magnet is removed.

(53) Then, the first membrane valve portion **132** is opened, and the air pressure is applied to the portion of the first sealing film **20** covering the extraction region **121** by the above-mentioned air pressure applying method, thereby driving the second rinsing liquid pre-embedded in the third liquid cell **143** to flow into the extraction region **121**. Next, the first membrane valve portion **132** is closed, and the high-frequency positive and negative air pressure is applied to the portion of the first sealing film **20** that covers the extraction region **121**, so that the portion of the first sealing film **20** that covers the extraction region **121** vibrates repeatedly, thereby allowing the liquid in the extraction region **121** to vibrate, so as to wash away salt ions and some small molecules. Then, the first membrane valve portion **134** is opened, and the magnet in the detection device is used to attract the magnetic beads **001** in the extraction region **121**. Air pressure is applied to the portion of the first sealing film **20** covering the extraction region **121**, so as to drive the liquid in the extraction region **121** to flow into the fifth liquid cell **145**. Then, the first membrane valve portion **134** is closed and the magnet is removed.

(54) Then, the first membrane valve portion **131** is opened, and the air pressure is applied to the portion of the first sealing film **20** covering the extraction region **121** by the above-mentioned air pressure applying method, so that the eluent pre-embedded in the fourth liquid cell **144** is injected into the extraction region **121**. The nucleic acid fragments adsorbed on the magnetic beads **001** are melted and eluted by the eluent, and are separated from the magnetic beads **001**. Next, the first membrane valve portion **131** is closed, and the second membrane valve portions **136** and **137** are opened. The air pressure is applied to the portion of the first sealing film **20** covering the extraction region **121** by the above-mentioned air pressure applying method, and the liquid containing the eluted nucleic acid fragments is injected into the reaction region **123**. At this time, the reaction region **123** is communicated with the fifth liquid cell **145** to use the fifth liquid cell **145** as a vent hole, so as to facilitate the liquid entering the reaction region **123**. In the process of driving the

liquid to flow into the reaction region **123**, the magnet in the detection device is used to attract the magnetic beads **001** in the extraction region **121**, so as to prevent the magnetic beads **001** from entering the reaction region **123**. Then the second membrane valve portions **136** and **137** are closed.

(55) Finally, the first membrane valve portion **134** is opened, and the magnet in the detection device is moved down to be away from the first sealing film **20**, so that the magnetic beads **001** can be moved and driven into the fifth liquid cell **145** along with the waste liquid. The amplification primer embedded in the liquid containing hole **002** of the reaction region **123** is melted by the liquid entering the liquid containing hole **002**. The temperature control unit in the detection device is used to control the temperature of the portion of the first sealing film **20** covering the reaction region **123**, so that the nucleic acid fragments in the reaction region **123** undergo constant temperature amplification or polymerase chain reaction (PCR). Then, the amplified product is analyzed and detected by the optical detection unit of the detection device, so as to complete the detection and obtain the detection result. In the case where the amplification primers embedded in the plurality of liquid containing holes **002** are different, multiple detection can be realized.

(56) Through the above steps, the detection chip **100** can be used to realize the analysis and detection of the sample to be detected. The detection chip **100** has a simple structure and a simple manufacturing process, and can improve the product yield, reduce the production cost, quantitatively deliver the reagent, achieve multiple detection, facilitate improving the heat conduction efficiency and the stability and accuracy of optical detection, and further effectively prevent leakage of the reagent during transportation.

(57) At least one embodiment of the present disclosure further provides a detection device, which is adapted to operate the detection chip according to any one of the embodiments of the present disclosure. The detection device can operate the aforementioned detection chip and can quantitatively deliver the reagent. Further, at least one example thereof can also realize multiple detection. Furthermore, at least one example thereof can also facilitate improving the heat conduction efficiency and the stability and accuracy of optical detection.

(58) FIG. 7A is a schematic block diagram of a detection device provided by at least one embodiment of the present disclosure. For example, as illustrated in FIG. 7A, a detection device **200** is adapted to operate the aforementioned detection chip **100**, and the detection device **200** includes a membrane valve control unit **210**.

(59) For example, the membrane valve control unit **210** is configured to mount the detection chip **100**, that is, after the membrane valve sealing plate **30** in the detection chip **100** is separated from the chip substrate **10**, the detection chip **100** can be mounted on the membrane valve control unit **210**. For example, in some examples, when the membrane valve sealing plate **30** is removed, the detection chip **100** is turned upside down, so as to prevent the liquid in each liquid cell **14** from flowing out of the liquid cell **14** after the membrane valve sealing plate **30** is removed. After the detection chip **100** is kept upside down and mounted on the membrane valve control unit **210**, the membrane valve portion **13** is closed, and then the detection chip **100** is turned over with the structure in contact with the membrane valve control unit **210**, so as to allow the detection chip **100** to be upright.

(60) FIG. 7B is a schematic structural diagram of a detection device provided by at least one embodiment of the present disclosure. For example, the detection device provided in this embodiment is basically the same as the detection device illustrated in FIG. 7A. For example, the membrane valve control unit **210** includes a main body portion **212** and at least one protruding portion **211** provided on the main body portion **212**. The main body portion **212** has a fixing structure for accommodating the above-mentioned detection chip **100**, for example, by means of clamping, bonding, etc., so as to fix the detection chip **100**. At least one protruding portion **211** can be moved (for example, protruding or retracting with respect to the main body portion **212**) to control the portion of the first sealing film **20** covering the membrane valve portion **13** whether to

approach the membrane valve portion **13** or whether to separate from the membrane valve portion **13** in the case where the detection chip **100** is mounted on the membrane valve control unit **210**, so as to enable the fluid channel **12** to be closed and opened correspondingly. For example, the protruding portion **211** may be driven by methods such as pneumatic, hydraulic or other suitable means, or the protruding portion **211** may be driven by a stepping motor, and these driving components are provided within the main body portion **212** of the membrane valve control unit **210**.

(61) For example, the function of the protruding portion **211** may be similar to the function of the aforementioned protruding structure **31**, that is, there may be a plurality of protruding portions **211** in one-to-one correspondence with the plurality of membrane valve portions **13**, so as to control the plurality of membrane valve portions **13** to be opened and closed, respectively. In the detection device **200**, the membrane valve control unit **210** can independently control each protruding portion **211**, and in the case where the protruding portion **211** and the corresponding membrane valve portion **13** are in contact with each other, the fluid channel **12** can be closed.

(62) For example, in some examples, there are seven protruding portions **211**, which correspond to the distribution positions of the seven membrane valve portions **131** to **137** in the detection chip **100** illustrated in FIG. 1 to FIG. 6. Each protruding portion **211** can be individually controlled, for example, can be moved upwards respectively, so that each protruding portion **211** can be inserted into each membrane valve portion **13** correspondingly, so as to allow the portion of the first sealing film **20** covering the membrane valve portion **13** to be squeezed and deformed to be completely attached to the chip substrate **10**, thereby allowing the fluid channel **12** to be closed. When the fluid channel **12** needs to be opened, the corresponding protruding portion **211** is moved down and away from the membrane valve portion **13**, and the deformation of the portion of the first sealing film **20** covering the membrane valve portion **13** is restored to be separated from the chip substrate **10**, so that the fluid channel **12** can be opened. For example, the protruding portion **211** may be a cylindrical protrusion, so as to cooperate with the circular membrane valve portion **13**.

(63) It should be noted that, in the embodiments of the present disclosure, as described above, the specific implementation of the membrane valve control unit **210** is not limited. For example, the membrane valve control unit **210** may be a combination of a hydraulic device, a propulsion control mechanism (such as a control circuit or a control chip), a cylinder (serving as the protruding portion **211**), and a position limiting mechanism, or may also be a combination of a motor, a propulsion control mechanism, a cylinder, and a position limiting mechanism, or in any other implementation manner, which can be determined according to actual requirements.

(64) FIG. 8 is a schematic block diagram of another detection device provided by at least one embodiment of the present disclosure. For example, as illustrated in FIG. 8, except for further including a film driving unit **220**, the detection device **200** provided in this embodiment is basically the same as the detection device **200** illustrated in FIG. 7A. In this embodiment, in the case where the fluid channel **12** of the detection chip **100** includes the extraction region **121**, the film driving unit **220** is configured to, in the case where the detection chip **100** is mounted on the membrane valve control unit **210**, apply pressure to the portion of the first sealing film **20** covering the extraction region **121** so as to allow the portion of the first sealing film **20** covering the extraction region **121** to deform, for example, to repeatedly vibrate. The portion of the first sealing film **20** covering the extraction region **121** is pushed up and down and squeezed, and thus in the case where the fluid channel **12** is not closed, the liquid can be pumped quantitatively, so as to control the flow of the liquid in the detection chip **100**.

(65) For example, in some examples, the film driving unit **220** may be an air pressure applying unit. The film driving unit **220** is configured to, in the case where the detection chip **100** is mounted on the membrane valve control unit **210**, apply air pressure to the portion of the first sealing film **20** covering the extraction region **121** of the fluid channel **12**. For example, the air pressure may be alternating positive air pressure and negative air pressure, or may be only positive air pressure or

only negative air pressure, which is not limited in the embodiments of the present disclosure. For example, the changing frequency of alternating positive air pressure and negative air pressure can be adjusted, so that a high frequency of changing air pressure and a low frequency of changing air pressure can be provided. The changing air pressure with the high frequency enables the liquid in the extraction region **121** to vibrate, so as to better perform operations such as extraction, rinsing, elution, etc. The changing air pressure with the low frequency can pump the liquid, so that the liquid can flow between the liquid cells **14**, the extraction region **121**, and the reaction region **123**. (66) It should be noted that in the embodiments of the present disclosure, the specific implementation manner of the film driving unit **220** is not limited. For example, the film driving unit **220** may be a combination of a pressure control device, an air compressor, and a gas delivery pipe, or may also be in any other implementation manner, which can be determined according to actual needs.

(67) It should be noted that, in the embodiments of the present disclosure, the detection device **200** may further include more components and units, which are not limited to the membrane valve control unit **210** and the film driving unit **220** described above. For example, the detection device **200** may also include a power supply, a central processing unit (CPU), an optical detection unit, a temperature control unit, etc., so that the detection device **200** may have complete and rich functions. The detailed description and technical effects of the detection device **200** may be referred to the above description of the detection chip **100**, and details are not described herein again.

(68) At least one embodiment of the present disclosure further provides a method for using a detection chip, and the method can be used to operate the detection chip described in any one of the embodiments of the present disclosure. By adopting this method, the reagent can be delivered quantitatively. Further, at least one example thereof can also realize multiple detection. Furthermore, at least one example thereof can also facilitate improving the heat conduction efficiency and the stability and accuracy of optical detection.

(69) FIG. **9** is a schematic flowchart of a method for using a detection chip provided by at least one embodiment of the present disclosure. For example, as illustrated in FIG. **9**, in some examples, the method includes following operations.

(70) Step **S00**: providing the detection chip **100**.

(71) Step **S10**: controlling the plurality of membrane valve portions **13** to allow the portion of the first sealing film **20** covering the plurality of membrane valve portions **13** to be separated from the plurality of membrane valve portions **13**, respectively, so as to open the fluid channel **12** correspondingly.

(72) FIG. **10** is a schematic flowchart of step **S10** of the method for using the detection chip illustrated in FIG. **9**. For example, as illustrated in FIG. **10**, in some examples, in the case where the plurality of membrane valve portions **13** include a plurality of first membrane valve portions **131** to **135** and a plurality of second membrane valve portions **136** to **137**, the chip substrate **10** includes a plurality of liquid cells **14**, and the fluid channel **12** includes an extraction region **121** and a reaction region **123**, the above step **S10** further includes following operations.

(73) Step **S110**: controlling the plurality of first membrane valve portions **131** to **135** to allow the plurality of liquid cells **14** to be communicated with the extraction region **121**, so as to allow liquid in the plurality of liquid cells **14** to enter the extraction region **121**.

(74) Step **S120**: controlling the plurality of second membrane valve portions **136** to **137** to allow the reaction region **123** to be communicated with the extraction region **121**, so as to allow liquid in the extraction region **121** to enter the reaction region **123**.

(75) For example, in step **S110**, by controlling the first membrane valve portions **131** to **135**, the plurality of liquid cells **14** can be sequentially communicated (for example, communicated in different operation phases, respectively) with the extraction region **121** of the fluid channel **12**, so that the liquid in the plurality of liquid cells **14** enters the extraction region **121** sequentially (for

example, enters the extraction region **121** in different operation phases, respectively).

(76) FIG. **11** is a schematic flowchart of another method for using the detection chip provided by at least one embodiment of the present disclosure.

(77) For example, as illustrated in FIG. **11**, in some examples, the method includes following operations.

(78) Step **S10**: controlling the plurality of membrane valve portions **13** to allow the portion of the first sealing film **20** covering the plurality of membrane valve portions **13** to be separated from the plurality of membrane valve portions **13**, respectively, so as to allow the fluid channel **12** to be opened correspondingly.

(79) Step **S20**: controlling flow of liquid in the detection chip **100** by applying pressure to the portion of the first sealing film **20** covering the extraction region **121**.

(80) For example, step **S10** in this embodiment is basically the same as step **S10** of the method illustrated in FIG. **9**, and details are not described herein again. For example, in step **S20**, the pressure may be alternating positive air pressure and negative air pressure, or may be only positive air pressure or only negative air pressure, which may be determined according to actual requirements, and the embodiments of the present disclosure are not limited in this aspect. For example, the changing frequency of alternating positive air pressure and negative air pressure can be adjusted, so that a high frequency of changing air pressure and a low frequency of changing air pressure can be provided. The changing air pressure with the high frequency enables the liquid in the extraction region **121** to vibrate, so as to better perform operations such as extraction, rinsing, elution, etc. The changing air pressure with the low frequency can pump the liquid, so that the liquid can flow between the liquid cells **14**, the extraction region **121**, and the reaction region **123**.

(81) It should be noted that, in the embodiments of the present disclosure, the method may further include more steps, which may be determined according to actual requirements, and the embodiments of the present disclosure are not limited in this aspect. The detailed description and technical effects of the method may be referred to the above description of the detection chip **100** and the detection device **200**, and details are not described herein again.

(82) The following statements should be noted. (1) The accompanying drawings involve only the structure(s) in connection with the embodiment(s) of the present disclosure, and other structure(s) can be referred to common design(s). (2) In case of no conflict, features in one embodiment or in different embodiments can be combined to obtain new embodiments.

(83) What have been described above are only specific implementations of the present disclosure, the protection scope of the present disclosure is not limited thereto, and the protection scope of the present disclosure should be based on the protection scope of the claims.

Claims

1. A detection chip, comprising: a chip substrate and a first sealing film that are stacked, wherein the chip substrate comprises a first surface, and the first sealing film covers the first surface of the chip substrate; the chip substrate further comprises a fluid channel on the first surface, and the fluid channel comprises a plurality of membrane valve portions; and the membrane valve portions are configured to allow a portion of the first sealing film covering the membrane valve portions to approach and separate, so as to close and open the fluid channel correspondingly, wherein the detection chip further comprises a membrane valve sealing plate, wherein the membrane valve sealing plate is on a side of the first sealing film away from the chip substrate and comprises a plurality of protruding structures, the plurality of protruding structures and the plurality of membrane valve portions are in one-to-one correspondence, and in a case where the plurality of protruding structures are in contact with the plurality of membrane valve portions, respectively, the fluid channel is closed, wherein the chip substrate further comprises at least one liquid cell, the at least one liquid cell is communicated with the fluid channel, at least one of the plurality of

membrane valve portions is configured to close and open a portion of the fluid channel communicated with the at least one liquid cell, wherein the fluid channel further comprises an extraction region and a plurality of first branches, and the at least one liquid cell comprises a plurality of liquid cells; the plurality of first branches are communicated with the plurality of liquid cells in one-to-one correspondence, and each of the plurality of first branches is communicated with the extraction region; the plurality of membrane valve portions comprise a plurality of first membrane valve portions respectively on the plurality of first branches, so as to control the plurality of first branches to be opened or closed; the plurality of liquid cells are distributed in a circular shape around the extraction region, wherein the fluid channel further comprises a reaction region and a plurality of second branches; the reaction region is communicated with the extraction region and at least one of the plurality of liquid cells through the plurality of second branches, respectively; the plurality of membrane valve portions further comprise a plurality of second membrane valve portions respectively on the plurality of second branches, so as to control the plurality of second branches to be opened or closed, wherein the reaction region comprises a porous structure, the porous structure comprises a plurality of liquid containing holes, and the plurality of liquid containing holes are configured to contain identical or different amplification primers, a position of one of the plurality of liquid containing holes serves as a center position, and remaining liquid containing holes among the plurality of liquid containing holes are distributed in a circular shape around the center position; the detection chip further comprises a second sealing film, wherein the chip substrate comprises a second surface opposite to the first surface, and the second sealing film covers the second surface of the chip substrate, wherein the second sealing film is a composite film comprising metal foil and a polymer material that are stacked.

2. The detection chip according to claim 1, wherein the first sealing film is an elastic film.

3. The detection chip according to claim 1, wherein the plurality of liquid cells comprise a first liquid cell, a second liquid cell, a third liquid cell, a fourth liquid cell, and a fifth liquid cell, the first liquid cell is configured to contain a lysis solution, the second liquid cell is configured to contain a first rinsing solution, the third liquid cell is configured to contain a second rinsing solution, the fourth liquid cell is configured to contain an eluent, and the fifth liquid cell is configured to accommodate a waste liquid generated in the extraction region during a reaction process.

4. The detection chip according to claim 1, further comprising an adhesive layer, wherein the adhesive layer is between the chip substrate and the first sealing film, and is configured to allow the chip substrate to be adhered with the first sealing film, and the adhesive layer exposes the fluid channel of the chip substrate.

5. A detection device, adapted to operate a detection chip, wherein the detection chip comprises: a chip substrate and a first sealing film that are stacked, the chip substrate comprises a first surface, and the first sealing film covers the first surface of the chip substrate; the chip substrate further comprises a fluid channel on the first surface, and the fluid channel comprises a plurality of membrane valve portions; and the membrane valve portions are configured to allow a portion of the first sealing film covering the membrane valve portions to approach and separate, so as to close and open the fluid channel correspondingly; the detection device comprises a membrane valve control unit, the membrane valve control unit is configured to mount the detection chip and comprises at least one protruding portion, and the at least one protruding portion is movable to control the portion of the first sealing film covering the membrane valve portions whether to approach the membrane valve portions or whether to separate from the membrane valve portions in a case where the detection chip is mounted on the membrane valve control unit, so as to close and open the fluid channel correspondingly, wherein the detection chip further comprises a membrane valve sealing plate, wherein the membrane valve sealing plate is on a side of the first sealing film away from the chip substrate and comprises a plurality of protruding structures, the plurality of protruding structures and the plurality of membrane valve portions are in one-to-one

correspondence, and in a case where the plurality of protruding structures are in contact with the plurality of membrane valve portions, respectively, the fluid channel is closed, wherein the chip substrate further comprises at least one liquid cell, the at least one liquid cell is communicated with the fluid channel, at least one of the plurality of membrane valve portions is configured to close and open a portion of the fluid channel communicated with the at least one liquid cell, wherein the fluid channel further comprises an extraction region and a plurality of first branches, and the at least one liquid cell comprises a plurality of liquid cells; the plurality of first branches are communicated with the plurality of liquid cells in one-to-one correspondence, and each of the plurality of first branches is communicated with the extraction region; the plurality of membrane valve portions comprise a plurality of first membrane valve portions respectively on the plurality of first branches, so as to control the plurality of first branches to be opened or closed; the plurality of liquid cells are distributed in a circular shape around the extraction region, wherein the fluid channel further comprises a reaction region and a plurality of second branches; the reaction region is communicated with the extraction region and at least one of the plurality of liquid cells through the plurality of second branches, respectively; the plurality of membrane valve portions further comprise a plurality of second membrane valve portions respectively on the plurality of second branches, so as to control the plurality of second branches to be opened or closed, wherein the reaction region comprises a porous structure, the porous structure comprises a plurality of liquid containing holes, and the plurality of liquid containing holes are configured to contain identical or different amplification primers, a position of one of the plurality of liquid containing holes serves as a center position, and remaining liquid containing holes among the plurality of liquid containing holes are distributed in a circular shape around the center position; the detection chip further comprises a second sealing film, wherein the chip substrate comprises a second surface opposite to the first surface, and the second sealing film covers the second surface of the chip substrate, wherein the second sealing film is a composite film comprising metal foil and a polymer material that are stacked.

6. The detection device according to claim 5, further comprising a film driving unit, wherein the film driving unit is configured to, in the case where the detection chip is mounted on the membrane valve control unit, apply pressure to a portion of the first sealing film covering the extraction region, so as to allow the portion of the first sealing film covering the extraction region to deform.

7. A method for using a detection chip, wherein the detection chip comprises: a chip substrate and a first sealing film that are stacked, the chip substrate comprises a first surface, and the first sealing film covers the first surface of the chip substrate; the chip substrate further comprises a fluid channel on the first surface, and the fluid channel comprises a plurality of membrane valve portions; and the membrane valve portions are configured to allow a portion of the first sealing film covering the membrane valve portions to approach and separate, so as to close and open the fluid channel correspondingly; the method comprises: controlling the plurality of membrane valve portions to allow the portion of the first sealing film covering the plurality of membrane valve portions to be separated from the plurality of membrane valve portions, respectively, so as to open the fluid channel correspondingly, wherein the detection chip further comprises a membrane valve sealing plate, wherein the membrane valve sealing plate is on a side of the first sealing film away from the chip substrate and comprises a plurality of protruding structures, the plurality of protruding structures and the plurality of membrane valve portions are in one-to-one correspondence, and in a case where the plurality of protruding structures are in contact with the plurality of membrane valve portions, respectively, the fluid channel is closed, wherein the chip substrate further comprises at least one liquid cell, the at least one liquid cell is communicated with the fluid channel, at least one of the plurality of membrane valve portions is configured to close and open a portion of the fluid channel communicated with the at least one liquid cell, wherein the fluid channel further comprises an extraction region and a plurality of first branches, and the at least one liquid cell comprises a plurality of liquid cells; the plurality of first branches are communicated

with the plurality of liquid cells in one-to-one correspondence, and each of the plurality of first branches is communicated with the extraction region; the plurality of membrane valve portions comprise a plurality of first membrane valve portions respectively on the plurality of first branches, so as to control the plurality of first branches to be opened or closed; the plurality of liquid cells are distributed in a circular shape around the extraction region, wherein the fluid channel further comprises a reaction region and a plurality of second branches; the reaction region is communicated with the extraction region and at least one of the plurality of liquid cells through the plurality of second branches, respectively; the plurality of membrane valve portions further comprise a plurality of second membrane valve portions respectively on the plurality of second branches, so as to control the plurality of second branches to be opened or closed, wherein the reaction region comprises a porous structure, the porous structure comprises a plurality of liquid containing holes, and the plurality of liquid containing holes are configured to contain identical or different amplification primers, a position of one of the plurality of liquid containing holes serves as a center position, and remaining liquid containing holes among the plurality of liquid containing holes are distributed in a circular shape around the center position; the detection chip further comprises a second sealing film, wherein the chip substrate comprises a second surface opposite to the first surface, and the second sealing film covers the second surface of the chip substrate, wherein the second sealing film is a composite film comprising metal foil and a polymer material that are stacked.

8. The method according to claim 7, wherein controlling the plurality of membrane valve portions to allow the portion of the first sealing film covering the plurality of membrane valve portions to be separated from the plurality of membrane valve portions, respectively, so as to open the fluid channel correspondingly, comprises: controlling the plurality of first membrane valve portions to allow the plurality of liquid cells to be communicated with the extraction region, so as to allow liquid in the plurality of liquid cells to enter the extraction region; and controlling the plurality of second membrane valve portions to allow the reaction region to be communicated with the extraction region, so as to allow liquid in the extraction region to enter the reaction region.

9. The method according to claim 8, further comprising: controlling flow of liquid in the detection chip by applying pressure to a portion of the first sealing film covering the extraction region.
